

```
In [3]: import pandas as pd
import time
import csv
import re
comp=pd.read_csv('glycaninsilico.csv', header=None)
sortcomp = comp.sort_values([1]).reset_index(drop=True)
```

```
In [8]: #testing block

print("Beofre sorting")
for i in range(3):
    print(comp[1][i])
print("After sorting")
for i in range(22):
    print(str(sortcomp[1][i]) + " mass <-, composition ->" + str(sortcomp[0][i]))
```

```
Beofre sorting
1149.601
1540.785
1931.97
After sorting
1149.601 mass <-, composition ->(0, 3, 2, 0, 0)
1323.69 mass <-, composition ->(1, 3, 2, 0, 0)
1353.701 mass <-, composition ->(0, 4, 2, 0, 0)
1394.727 mass <-, composition ->(0, 3, 3, 0, 0)
1497.78 mass <-, composition ->(2, 3, 2, 0, 0)
1510.775 mass <-, composition ->(0, 3, 2, 1, 0)
1527.79 mass <-, composition ->(1, 4, 2, 0, 0)
1540.785 mass <-, composition ->(0, 3, 2, 0, 1)
1557.801 mass <-, composition ->(0, 5, 2, 0, 0)
1568.817 mass <-, composition ->(1, 3, 3, 0, 0)
1598.827 mass <-, composition ->(0, 4, 3, 0, 0)
1639.854 mass <-, composition ->(0, 3, 4, 0, 0)
1671.869 mass <-, composition ->(3, 3, 2, 0, 0)
1684.864 mass <-, composition ->(1, 3, 2, 1, 0)
1701.870 mass <-, composition ->(0, 4, 2, 0, 0)
```

```

In [3]: #modified from programiz
def binarycompositionsearch(array, x, low, high, boundary):
    # Repeat until the pointers low and high meet each other
    while low <= high:
        mid = low + (high - low)//2
        #print(f"mid is {mid} now")
        #print(f"value: {array[mid]-x}")
        if abs((array[mid])-x) < boundary:
            return mid
        elif ((array[mid])-x) < -(boundary):
            #print("mid<x")
            low = mid + 1
            #print(f"now lower bound is {low} and it's {array[low]}")
        else:
            #print("mid>x")
            high = mid - 1
            #print(f"now higher bound is {high} and it's {array[high]}")
    return -1

array = sortcomp[1]
callcomp = sortcomp[0]

x = 1714
boundary = 2
result = binarycompositionsearch(array, x, 0, len(array)-1, boundary)

#Waiting: Extend search to find out all
#the index has a mass difference less than defined BOUNDARY

if result != -1:
    print("Element is present at index " + str(result))
    #go upper
    upsearch = result+1
    lowsearch = result-1
    if abs((array[upsearch])-x) < boundary:
        print("Element is ALSO present upper at index " + str(upsearch) + ":" + str(sortcomp[upsearch]))
        upsearch = result+1
    else:
        print("Stop expanding searching upper element")
    if abs((array[lowsearch])-x) < boundary:
        print("Element is present lower at index " + str(lowsearch) + ":" + str(sortcomp[lowsearch]))
        lowsearch = result-1
    else:
        print("Stop expanding searching lower element")
else:
    print("Not found")

```

Element is present at index 15

Element is ALSO present upper at index 16:1714.875

Stop expanding searching lower element

```
In [4]: print(f"{array[15]}+{callcomp[15]}")
```

1714.875+(1, 3, 2, 0, 1)

```
In [5]: print(f"{array[16]}+{callcomp[16]}")
```

1714.875+(0, 4, 2, 1, 0)

```

In [9]: #this blocok is designed for looping through the spectra
def compileresult(result, boundary):
    foundindex = []
    interpretedmz = None
    interpretedcomp = None
    upper = True
    lower = True
    #print(f"result is {result}")
    if result != -1:
        interpretedmz = array[result]
        interpretedcomp = callcomp[result]
        foundindex.append((interpretedmz, interpretedcomp))
        interpretedmz = None
        interpretedcomp = None
        upsearch = result+1
        escapelower = False
        if result > 0:
            lowsearch = result-1
        else:
            lowsearch = 0
            escapelower = True
        print(foundindex)
        while abs((array[upsearch]-array[result]) < boundary:
            #print("upper test")
            #print(f"upsearch{upsearch}")
            interpretedmz = array[upsearch]
            interpretedcomp = callcomp[upsearch]
            foundindex.append((interpretedmz, interpretedcomp))
            #print(f"debug mass diff{(array[upsearch]-array[result])} and index {upsearch}")
            upsearch = upsearch+1
            #print(f"now the index is {upsearch}")
        #print("Stop expanding searching upper element")
        if escapelower is True:
            print("Trying to search a mass lower than in silico minimum")
            while abs((array[lowsearch]-array[result]) < boundary and not escapelower:
                #print("lower test")
                #print(f"lowsearch{lowsearch}")
                interpretedmz = array[lowsearch]
                interpretedcomp = callcomp[lowsearch]
                foundindex.append((interpretedmz, interpretedcomp))
                lowsearch = lowsearch-1
                #print(f"debug mass diff{(array[upsearch]-array[result])}")
                #print(f"now the index is {lowsearch}")
            #print("Stop expanding searching lower element")
            #print(f"Found index: {foundindex}")
            return foundindex
        else:
            #print("None")
            return foundindex

result = binarycompositionsearch(array, x, 0, len(array)-1, boundary)
compileresult(result, boundary)

```

```
[(1714.875, ' (1, 3, 2, 0, 1)')]
```

```
Out[9]: [(1714.875, ' (1, 3, 2, 0, 1)'), (1714.875, ' (0, 4, 2, 1, 0)')]
```

```
In [28]: #read the file
searchinput = pd.read_csv('zf_sPerMeNG_brain MS2 summary from 44218 at 20230307-153550.c
searchinputindex = searchinput["in [H+]"]
#set boundary
boundary = 2
foundindextmp = []
for i in range(100):
    print(f"mz from spectra file {searchinputindex[i]}")
    tmp = int(searchinputindex[i])
    result = binarycompositionsearch(array, tmp, 0, len(array)-1, boundary)
    compileresult(result, boundary)
    print("-"*10)
```

mz from spectra file 1026.795138515625

None

mz from spectra file 1118.80612484375

None

mz from spectra file 1118.806368984375

None

mz from spectra file 1046.7845183984375

None

mz from spectra file 1046.7847625390625

None

mz from spectra file 1100.7962371484375

None

mz from spectra file 1118.806857265625

None

```
In [10]: #try to add the returned value into df
searchinput = pd.read_csv('zf_sPerMeNG_brain MS2 summary from 44218 at 20230307-153550.c
searchinputindex = searchinput["in [H+]"]
#set boundary
boundary = 2
foundindextmp = []
for i in range(len(searchinput)):
    print(f"mz from spectra file {searchinputindex[i]}")
    tmp = int(searchinputindex[i])
    result = binarycompositionsearch(array, tmp, 0, len(array)-1, boundary)
    foundindextmp.append(compileresult(result, boundary))
searchinput['Predicted composition'] = foundindextmp

searchinput.head(100)
```

```
mz from spectra file 1026.795138515625
mz from spectra file 1118.80612484375
mz from spectra file 1118.806368984375
mz from spectra file 1046.7845183984375
mz from spectra file 1046.7847625390625
mz from spectra file 1100.7962371484375
mz from spectra file 1118.806857265625
mz from spectra file 1054.7900115625
mz from spectra file 1118.8064910546875
mz from spectra file 1078.75632015625
mz from spectra file 1100.79635921875
mz from spectra file 1178.8396941796875
mz from spectra file 1100.7959930078125
mz from spectra file 1118.80612484375
mz from spectra file 1178.8384734765625
mz from spectra file 1100.7959930078125
mz from spectra file 1078.7568084375
mz from spectra file 1138.833224453125
mz from spectra file 1118.8064910546875
```

```
In [11]: #save the file into another file and generate a setting file... does setting file needed
timestamp = time.strftime("%Y%m%d-%H%M%S") #datetime.datetime.now().strftime("%Y%m%d-%H%M%S")
extfilename= "zf_sPerMeNG_brain MS2 summary from 44218 at 20230307-153550 " + timestamp
searchinput.to_csv(extfilename)
print("export finished") #Doesn't check the existence of file, should be modified in the
```

```
In [1]: #find the folder contains *predictcomp.csv
import fnmatch
import os

for file in os.listdir('.'):
    if fnmatch.fnmatch(file, '*predictedcomp.csv'):
        print(file)
#https://docs.python.org/3/library/fnmatch.html#module-fnmatch
```

```
zf_sPerMeNG_brain MS2 summary from 44218 at 20230307-153550 20230507-181703 predicted
comp.csv
```

In []: